HIKET — Materials and methods

complete version for review, to be merged into the second draft

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1 Materials and methods

The section first describes the data (the soil carbon observations, the litter input and the climate), then the six models and how they are started and calibrated, and then the calibrations and simulations built on that common setup: the comparison with a soil started in balance, the division of the modelled sink among its sources, and the simulations beyond the observations. The evaluation of the models against the observations closes the section.

1.1 Study plots and soil carbon observations

The study is based on data from permanent forest plots of the Finnish National Forest Inventory for which soil carbon has been measured in three campaigns: the eighth national forest inventory (VMI8), the BioSoil campaign of 2006, and the Komeetta campaign of 2024. After the exclusions described below, 456 plots are available for calibration, carrying 1205 plot-year observations of soil carbon in total. Of the 456 plots, 365 plots (80 per cent, carrying 959 observations) were used to fit the models, and 91 plots (20 per cent, carrying 246 observations) were held out and used only for evaluation. Plots were assigned to the two groups at random, and a plot was assigned as a whole, so that no plot contributes observations to both groups.

The first campaign is not a single year. Its plots were sampled region by region between 1986 and 1995: 69 plots in 1986, 76 in 1987, 103 in 1988, 54 in 1989 and 72 in 1995. Averaged over the plots, the sampling year of the first campaign is 1989, and that is the year at which the campaign is drawn whenever it is shown as a single point. In the calibration, each observation is compared with the model in the year in which that plot was actually sampled.

On the plots measured in all three campaigns (310 plots, the set on which every observed rate of change in this paper is computed), the mean stock of the groups sampled in 1986, 1987, 1988, 1989 and 1995 differs by 18.3 tonnes of carbon per hectare at first visit, which might reflect a trend within that same campaign. The five groups of plots defined by their first-campaign sampling year differ by 37.2 tonnes in 2006 and by 34.2 tonnes in 2024, years in which every plot of every group was measured in the same campaign. The difference between the five groups is therefore a property of which plots each group contains, and not of the year in which it was sampled, because the difference is larger when the sampling year is held constant.

[The exclusion rules (zero litter, absent organic layer, peat soils, constant litter series, whole-profile stock above 250 t C ha^{-1}) are referred to as "described below" but are not yet written out; add one short paragraph listing them with the number of plots each removes.]

1.2 Soil carbon stocks on a single homogenised basis

Since the three campaigns were measured and processed under protocols that differed in ways that matter for a comparison between them, all three campaigns were rebuilt on a uniform basis, starting from the layer-by-layer carbon stocks measured in each of the three campaigns and applying one set of conventions to all three. Those stocks are corrected for the volume occupied by stones and for the bulk density of the fine earth. The volume of stones was measured in 2006 on every plot by the rod-penetration method of Viro (1952), in which a steel rod is driven into the mineral soil at the sampling points and the mean depth it reaches is converted into a stone volume, and the same stone volume is applied to the first campaign (Lehtonen et al., 2016; Kramarenko, 2012). The bulk density of the mineral layers was predicted from the equations of Tamminen and Starr (1994) for all layers of the first campaign and for most layers of 2006, and measured directly for part of the 2006 samples from the upper 20 cm (Kramarenko, 2012). [Confirm that the 2024 campaign uses the same stone volumes and the same bulk density equations.] The coarse-fragment correction matters because the coarse-fragment content of these plots has a median of 43 per cent by volume, so a mineral stock computed without it is too large by a factor of about 1.6.

The quantity used as the calibration target is the whole soil profile: the measured organic layer, plus the measured mineral soil to 40 cm, plus a modelled deep tail, integrated only as far down as the auger actually reached on that plot and in no case deeper than one metre. The depth to which the profile is integrated is therefore a per-plot quantity and not a fixed one metre, since using a fixed depth of, for example, one metre would invent soil carbon below bedrock on the 38 plots where augering stopped at 10, 20 or 40 cm. The modelled deep tail was validated against the measured 40–80 cm layer of the BioSoil campaign, which was held out of the fit: the median ratio of predicted to observed carbon in that layer is 0.96, with a bias of −2.5 tonnes of carbon per hectare.

The rebuilt target was checked against the national figures published for the two campaigns for which such figures exist. Applying the same plot exclusions as the official calculation, and integrating over the same layers, the rebuilt target returns weighted national means of 59.06 and 61.05 tonnes of carbon per hectare for 2006 and 2024 against official values of 59.06 and 61.05, that is, agreement to the nearest 0.01 tonnes per hectare on 446 plots.

Two further steps were taken with the first campaign, both of them established from the structure of the measurements themselves rather than from the fit. First, the layers of the three campaigns were standardised, because they were recorded differently: the organic layer of the first campaign contains the fermentation and humus horizons only, whereas the organic layer of the two later campaigns also contains the litter and moss layer. Each plot's own litter and moss layer from 2006 was therefore added to its first-campaign organic layer, so that the same compartment is compared across the three campaigns. Second, the first campaign is treated as what it is, a campaign whose fieldwork ran over several years rather than a campaign carried out in a single year. The year in which each plot was actually sampled is on record, and those years, rather than one nominal year for the whole campaign, are the years at which the models are compared with the observations.

On this final basis the median whole-profile stock is 62.0 tonnes of carbon per hectare for the first campaign, 66.4 for 2006 and 67.2 for 2024, and the observations vary between plots with a coefficient of variation of 0.44 to 0.53 depending on the campaign. [Supplementary figure: F2_baseline]

[Data availability and acknowledgement: the soil carbon measurements are not a published dataset, so the statement has to say where they come from, who compiled them and what of the compiled form can be deposited. Settle this with the people who hold them before submission.]

1.3 Litter input

Litter input is taken from the annual reconstruction of tree biomass and litter input for the permanent plots of the Finnish National Forest Inventory by Ťupek et al. (2026), published as an open dataset. In that reconstruction, the tree biomass measured by the inventory is interpolated to every year with the help of satellite-derived basal area, and litter is computed as the biomass of each tree component multiplied by a turnover rate for that component, plus the branches, foliage, stumps and roots left on site when a drop in basal area identifies a harvest; the full method is given in Ťupek et al. (2026). The product supplies, for each plot and year, the carbon in litter from trees, separated into non-woody litter (foliage and fine roots), fine woody litter (branches and coarse roots) and coarse woody litter (stumps), and each of the three classes is further divided into the four chemical fractions used by the Yasso models, using proportions specific to each tree species and component. The product does not contain a separate term for trees that die without being harvested, and it does not include the understorey vegetation, because the product is a tree litter product by design. Across the study plots the mean litter input carried by the product is $\bar{J} = 2.511$ tonnes of carbon per hectare per year.

The national greenhouse gas inventory estimates the same flux for the mineral forest soils of southern and northern Finland, from the forest inventory and biomass models (Statistics Finland, 2025). On the part that the two estimates have in common, the litter from the turnover of living trees and from harvest residues, they agree in level in southern Finland, where most of the study plots lie, and neither rises after about 2006; in northern Finland the inventory estimate lies higher and keeps rising [Supplementary figure: S18_litter_vs_inventory]. The inventory estimate also carries the litter of trees that die without being harvested, a term that grows over the period and that the plot product does not contain. [confirm with LUKE that the two regions of the inventory data match the region code of the study plots]

The first year of the product is treated separately. The median litter input of 1985 is 0.060 tonnes of carbon per hectare per year against 1.398 in 1986, a step that the series does not repeat in any later year. Because 1985 is both the first year of the simulation and the year in which the first soil observations fall, the value for 1985 was reconstructed for each plot and each chemical fraction by fitting a linear trend over 1986–1990 and extrapolating one year backwards. The reconstruction places 1985 just below 1986, which is the direction the growing-stock record requires.

1.4 Climate

Climate forcing comes from the gridded daily weather observations of the Finnish Meteorological Institute, interpolated from the station network onto a 10 by 10 km grid by kriging with external drift (Aalto et al., 2016) and extended by the Institute to the present. The grids were obtained through the internal weather database of the Natural Resources Institute Finland, which mirrors them from the Finnish Meteorological Institute. From the grid cell nearest to each plot we took the daily mean, minimum and maximum temperature, precipitation and global radiation for 1961–2025, and each model takes from those series the variables it requires. The record is used in two ways. From 1985 onwards each year of the simulation is driven by the climate observed in that year. Before 1985 the simulation is driven by one constant climate, the mean of 1961–1984 at each plot (a plot mean of 2.2 degrees Celsius, against 2.7 for 1985–2004), so that the soil enters 1985 adjusted to the climate it had lived in; the same long-term mean serves as the reference climate wherever one is

needed later (Sections 1.7, 1.13 and 1.14).

Because the response of decomposition to climate is not a straight line, the average of the response over the years is not the response evaluated at the average climate. Averaging the climate first and evaluating the response afterwards biases the climate modifier by 5 to 15 per cent in boreal conditions. The climate modifier is therefore computed for each year and averaged afterwards, in all six models.

1.5 The six models

Six models are used, and their roles are different by class. Yasso is the model this study is focused on, because Yasso is the model the Finnish greenhouse gas inventory runs. All three published versions of Yasso are included: Yasso07, Yasso15 and Yasso20. Alongside the three versions of Yasso we included three simple models with one, two and three carbon pools, denoted SP1, TP2 and TP3. The three simple models are there as a benchmark, since their behaviour is transparent enough so that any result obtained with Yasso can be checked against a structure simple enough to reason about quickly.

Due to their different structures and designs, the number of free parameters differs greatly across the six models, from six free parameters in SP1 to twenty-six in Yasso15 and Yasso20. The free parameters of the three versions of Yasso are the twelve fractions describing transfers between pools, the climate parameters, the parameters describing the effect of the size of woody litter, and the two auxiliary parameters described in Section 1.6.

The decomposition rates of Yasso are not fitted, and are held at their published values. Holding the rates fixed is a deliberate choice to keep the model anchored to its original definitions. Decomposition rates and transfer fractions do not act independently on the carbon stock but are always interacting: a slower rate and a larger transfer into a slow pool produce much the same trajectory, so fitting both together improves usually little and makes the result harder to read.

Of the two, the rates are the more fundamental to the definition of each model, because the rate of a pool is what distinguishes that pool from the others. Whether the pools of a soil carbon model can also be identified with measurable fractions of organic matter is a long-standing question in soil carbon modelling, and the answer has shaped how Yasso was built. The first version of Yasso assumed that litter consists of groups of organic compounds, each group decomposing at its own rate regardless of the litter it came from, and defined its decomposition compartments accordingly as extractives, celluloses and lignin-like compounds, followed by two compartments of humus (Liski et al., 2005). Yasso07 kept the same principle and defined its four litter pools by fractions that can be measured on litter by sequential extraction: the fraction soluble in acid, the fraction soluble in water, the fraction soluble in ethanol, and the insoluble remainder (Tuomi et al., 2009). Defining the pools by the chemistry of the litter is what makes Yasso operational, because the chemistry of litter can be measured and supplied to the model directly. The same aim, of matching the pools of a model with fractions that can be isolated in the laboratory, has been pursued for other models, for example through fractionation schemes designed to correspond to the pools of RothC (Zimmermann et al., 2007), and whether fractions isolated in the laboratory correspond to the functional pools of models has been examined at length (von Lützow et al., 2007).

Once carbon has entered the soil, the parameterisation of Yasso moves carbon between the four chemical fractions in every direction. In the published parameterisation of Yasso07, for example, 97 per cent of the carbon decomposed from the insoluble fraction is passed to the acid-soluble

fraction, and 99 per cent of the carbon decomposed from the acid-soluble fraction is passed to the water-soluble fraction (Tuomi et al., 2009), which represents the transformation of organic matter by decomposers rather than the decay of a fixed set of compounds. The pools therefore enter the model defined by the chemistry of the litter, and within the soil they act, as in any other compartmental soil carbon model, as compartments defined by their rates of turnover.

The view of pools as compartments of turnover and the view of soil organic matter as a continuum of quality are less opposed than they appear. A chemical fractionation is itself a discretization: it divides a continuum into a small number of classes, the discretization is made in the laboratory rather than in the model, and where the boundaries between classes fall is set by the method chosen, so that the fractions are defined by the procedure that isolates them (von Lützow et al., 2007). The material being divided is not a set of discrete compounds. Soil organic matter is the product of many processes of decomposition, microbial transformation and recombination, much of that organic matter has no defined molecular structure, and the chemical diversity of soil organic matter is large enough that the diversity can be treated as a continuum (Lehmann and Kleber, 2015).

The theory of continuous quality starts from that continuum: Bosatta and Ågren (1991) describe soil organic matter by a distribution of quality, in which the quality of a unit of carbon determines how fast that unit is decomposed, and decomposition in turn moves the carbon along the continuum (Ågren and Bosatta, 1998). The same idea was formalised independently in the geochemistry of sediments, where models with a few discrete fractions of different reactivity are treated as discretizations of a continuous distribution of reactivity (Boudreau and Ruddick, 1991). On this view, a pool of a compartmental model and a fraction of a laboratory procedure are two discretizations of the same continuum, one made in the model and one made in the laboratory, and in the model the rate of decomposition is what places each pool on the continuum. Changing a rate therefore redefines the pool, whereas the transfer fractions describe how carbon moves between pools that have already been defined. We therefore keep the rates as the published parameterisation set them and calibrate the transfer fractions. In all three versions of Yasso the decomposition rates, together with the fraction of decomposed carbon that enters the humus pool, are fixed at their published values: they are constants of the model and not parameters of the calibration, and no prior is placed on them.

1.6 The uncertainty of the litter input

The litter input is among the least certain of the quantities that drive a soil carbon model. Litter is not measured on the plots but derived from tree biomass through biomass models and turnover rates, parts of the input such as the litter of the understorey are missing from tree-based products by construction, and the input carries year-to-year variation that periodic inventories cannot capture. Uncertainty analyses of the Finnish soil carbon estimate have identified the litter input, together with the initial state of the soil, as the dominant sources of uncertainty (Peltoniemi et al., 2006; Lehtonen and Heikkinen, 2016). Those analyses propagated the uncertainty forward, by sampling the input from assumed error distributions and running the soil model with each sample. We instead estimate the error of the input from the soil observations themselves, together with the parameters of the models, so that the correction applied to the input and the uncertainty of that correction are both learned from the data and carried into every result.

The correction is a single multiplier, σ_{input}, applied to the litter input of the product over the

whole simulation. The prior of σ_{input} is log-normal with a centre of 1.08, the ratio of the total litter input to Finnish upland soils estimated by Lehtonen and Heikkinen (2016), which includes the understorey, to the tree litter input of the product used here, 2.70 against 2.511 tonnes of carbon per hectare per year, and a width of 0.25 on the logarithmic scale. Because the multiplier is one number for the whole country, it corrects the national level of the input and not the input of individual plots, whose errors are carried by the error model described below. [The sensitivity to the centre of this prior (centre 1.27, the ratio implied by the reconstruction of Liski et al. (2006)) was run on an earlier version of the setup and not repeated on the present one: repeat it with the new input data, or drop the sentence.]

The same reasoning is extended to the starting state. The litter input of 1917, which sets the state of the soil at the start of the simulation, is not known either, and is estimated through a second parameter, σ_{init}, the ratio of the litter input of 1917 to the litter input of 1985. The prior of σ_{init} is log-normal with a centre of 0.90, the ratio of the growing stock of Finnish forests in 1917 to that in 1985 (Korhonen et al., 2024) converted to a ratio of litter input with the relation between litter input and growing stock observed in the product, and the same width of 0.25; the reconstruction of Liski et al. (2006), based on the same growing-stock record, implies a similar ratio, about 0.86. Both parameters are estimated together with the model parameters, so that the uncertainty of the input and of the starting state propagates into the initial state, into the simulated trajectory and into the predictive intervals of every model.

1.7 Transient initialisation

Rather than starting each plot from a soil in balance with its present litter input, each draw of the parameter vector starts the soil in 1917 and runs it forward to the first soil observation before the comparison with data begins. The year 1917 is a convention, chosen as the year of Finnish independence and therefore the point from which the national forest record is usually counted (any year starting far enough before the observations for the soil to carry a history would work). The state of the soil in 1917 is obtained conventionally, with a steady state assumption with a constant litter input equal to the litter input of 1917 and the constant climate of 1961–1984 (Section 1.4). The litter input is then raised along a prescribed shape over the sixty-eight years from 1917 to 1985, still under the same constant climate, after which the simulation is driven by the annual litter and climate of the record. A constant climate is used before 1985 because the record does not reach back to 1917; the mean of the earliest observed decades is the closest available measure of the climate in which the soil developed before the observations began.

The shape along which the litter input is raised is taken from the reconstruction of the litter input to Finland's forest soils published by Liski et al. (2006), who estimated that input annually from 1922 onwards by combining national forest inventory data with biomass and turnover models. Three decisions were made about that reconstruction. First, only the tree-derived terms are used, meaning tree litter together with harvest residues and the litter of trees that died naturally, because those are the terms closest to the litter product used after 1985; that product represents natural mortality only when a death appears as a loss of basal area, but natural mortality is between 2 and 4 per cent of the tree-derived input in the reconstruction, and leaving it out changes the shape used here by less than 0.04 in any year. The ground vegetation is left out of the shape for the same reason that it is left out after 1985, assuming that it is absorbed by the input multiplier described in Section 1.6, and including it before 1985 but not after would make that multiplier mean two different things on either side of the join. Second, the national totals are converted to a per-hectare basis

by dividing by the area of upland forest soil. Liski et al. (2006) report the carbon of upland soils both as a national total and per unit area, and the ratio of the two gives an area of 13.90 million hectares in 1922 and 15.22 million hectares in 2004. We placed the growth between the two values in 1965–1980, the period in which Korhonen et al. (2024) locate most of the expansion of productive forest land. Third, only the shape is taken and not the level: the series is rescaled to run from zero in 1917 to one in 1985, so that the reconstruction supplies only the path between the two endpoints. The two endpoints are set by the two parameters described in Section 1.6: the input at the end of the path is the litter input of the product, averaged over 1985–1989 for each plot and scaled by σ_{input}, and the input at the start of the path is a fraction σ_{init} of the input at the end.

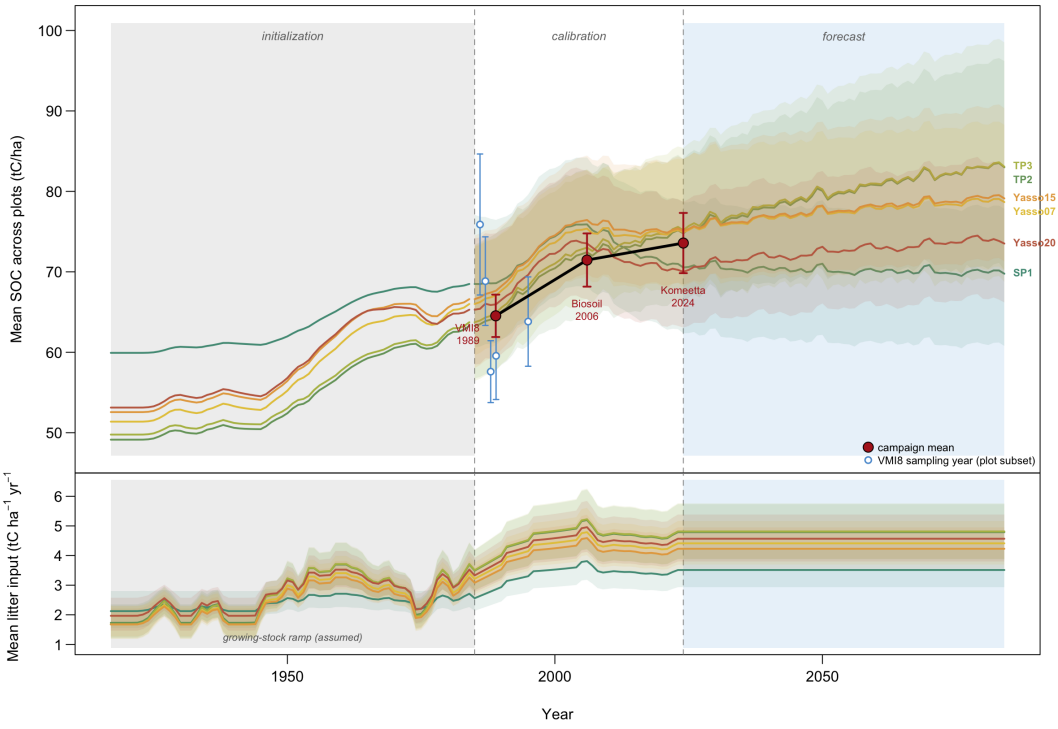


Figure 1: How the starting state of the soil is obtained. Rather than starting each plot from a soil already in balance with its present litter input, the simulation starts in 1917 from a soil in balance with the litter input of 1917 and the mean climate of 1961–1984, and the litter input is then raised over the following sixty-eight years along a shape taken from the reconstruction of Finland's forest litter input by Liski et al. (2006), after which the simulation is driven by the annual litter and climate of the record. The litter input of 1917 is not known and is estimated as part of the calibration.

1.8 Keeping the models as comparable as possible

The six models cannot be given the same external information. The models differ in structure and in the number of their parameters, and the published literature provides different information for each of them: the three versions of Yasso come with their own calibrations, whereas the three simple models have no calibration for forest soils. The more the models differ in the information they are given, in their assumptions and in their freedom, the less a difference between their results can be attributed to their structure. We therefore reduced that difference as far as the models allow,

through one compromise for each class of parameter, summarised in Table 1. The compromises reduce the asymmetry between the models without removing it, and what remains is stated with each compromise.

1.8.1 Decomposition rates

The rates of the three versions of Yasso are fixed at their published values, for the reasons given in Section 1.5. The three simple models have no rates of their own for forest soils, and take theirs from the ICBM model (Andrén and Kätterer, 1997): the fast rate is fixed, and the slow rate or rates are given a very informative prior centred on the ICBM value, so that the data can move them only slightly. The slow rates are pinned rather than fixed because the ICBM values cannot be transferred to a forest soil unchanged. ICBM was calibrated on an arable long-term experiment, and cropland and forest soils differ in what controls decomposition: cultivation breaks up the soil aggregates that protect organic matter physically in undisturbed soils (Six et al., 2002). Moreover, a rate is only meaningful together with the climate response it was estimated with, and the simple models combine the rates of ICBM with the climate response of Yasso07; the rates were rescaled once so that the two climate responses agree at the climate of the experiment on which ICBM was calibrated, but the two responses differ elsewhere. Fitting the rates freely, on the other hand, would give the simple models a freedom that the versions of Yasso do not have, and a narrow prior around the ICBM value is the compromise between the two.

1.8.2 Transfer fractions

The fractions that describe how carbon passes between pools are free in every model that has them: twelve in each version of Yasso, and the fraction of decomposed carbon passed on to the slower pool in the simple models, of which SP1 has none, TP2 one and TP3 two. All fractions carry a prior of the same width, 0.4 on the logit scale, centred on the published value of each model, which for the simple models is the humification coefficient of ICBM.

1.8.3 Climate response and size of woody litter

The three versions of Yasso take the centres and widths of their climate priors from their own published calibrations (Tuomi et al., 2009, 2011), and [Yasso15/20 source: Yasso20 = Viskari et al. 2022; Yasso15's calibration is unpublished (Järvenpää et al., in preparation) – decide whether to cite the FMI parameter files or the manuscript]. The simple models, which have no published climate response of their own, borrow the climate response of Yasso07, centre and width. Only the versions of Yasso describe how decomposition depends on the size of woody litter, with priors taken from their own calibrations; the simple models have no such term.

1.8.4 The two auxiliary parameters

Every model carries the two parameters described in Section 1.6, σ_{input} and σ_{init}, with the same prior, the same centre and the same width, in every model.

Table 1: How the prior of each class of parameter was set.

[Parameter-class by prior-criteria table: for each class (decomposition rates, transfer fractions, climate response, size of woody litter, the two auxiliary parameters), the source of the centre, the source of the width, the transformation and constraint, and which models carry it. The worked version is in `Prior_specs/PRIOR_HOMOGENIZATION_PLAN.md`, section 2.]

1.9 Numerical integration

Each model is run forward one year at a time, from 1917 to the last observation, with the litter input and the climate of each year held constant within that year. Within a year, each pool receives its share of the litter input and of the carbon passed on from other pools, and loses a fixed fraction of its own carbon per unit of time. The carbon in the pools at the end of the year can therefore be obtained in two ways. The simple way is to compute how fast each pool is changing at the start of the year and to assume that the pool keeps changing at that speed for the whole year. The exact way is to use the solution of the equations over the year, which for a single pool is known in closed form: the pool approaches the value at which gains and losses balance along an exponential curve, and never passes that value. No numerical solver of differential equations is used: within each year the equations of all six models have a known exact solution, and that solution is evaluated once per year. The only approximation is the annual time step itself, and re-running the calibrated three-pool model with twelve monthly steps a year changed the national mean stock by at most 0.7 tonnes of carbon per hectare in any year, about one per cent of the stock.

The two ways differ when a pool turns over faster than once a year. The simple way then carries the pool past its balance value in one year and back in the next, and produces an oscillation that belongs to the method of calculation and not to the model. The fastest pool of the three-pool model turns over more than once a year in warm years, so the difference matters here.

All six models use the exact solution. For several pools that pass carbon to one another the exact solution involves the exponential of the matrix of rates and transfers, and the models differ only in how that exponential is obtained. For the one-pool and two-pool models the solution is written out as a formula. For the three-pool model carbon moves in one direction only, from the fast pool to the slow pool and from the slow pool to the humus, so the solution can also be written out as a formula, as a combination of one exponential decay per pool. In the three versions of Yasso carbon moves between pools in every direction, and the exponential is computed numerically in the Fortran code of the models by the standard method of scaling and squaring (Moler and Van Loan, 2003).

1.10 Mean transit time as a common measure of a model

The six models differ in how many carbon pools they contain and in how carbon moves between those pools, so their parameters cannot be compared with one another directly. A single number can be compared, however, and that number is the mean time a unit of carbon spends in the soil between entering it as litter and leaving it as respired carbon dioxide. Sierra et al. (2017) call that quantity the *mean transit time*, and we use their term throughout. Much of the soil carbon literature, including earlier work of our own, calls the same quantity the mean residence time; Sierra and coauthors show that the term residence time has been used for at least three different quantities and discourage its use, so we adopted their convention. At steady state the mean transit time is also equal to the total carbon stock divided by the total input flux, which is what many authors call

the turnover time.

For a system in which each pool loses carbon in proportion to the amount of carbon it holds, and in which the rates do not change with time, the mean transit time can be computed from the rates and transfers of the model alone (Rasmussen et al., 2016; Sierra et al., 2017): at steady state the mean transit time equals the total carbon the system holds divided by the flux of carbon entering it. We therefore give each model a litter input of exactly one tonne of carbon per hectare per year and compute the carbon stock at which the gains and losses of every pool balance. The balance stock is computed directly and not by running the model until it settles: for the three simple models the balance stock is written out as a formula, and for the three versions of Yasso the balance equations of all pools are solved together as one linear system. The mean transit time is then the balance stock divided by the input. Because the models are linear, the result does not depend on the amount of input chosen: doubling the input doubles the stock and leaves the ratio unchanged. An input of one tonne only makes the stock and the mean transit time the same number; the quantity reported is the ratio, in years.

Two properties make the quantity useful as a basis for comparing models. First, the mean transit time computed in this way is a property of each model alone. Neither the litter input multiplier, nor the estimated starting state, nor any soil carbon observation enters the calculation, so two models can be compared even though they were fitted to the same data with different numbers of parameters, and our calibrated parameters can be compared with the published parameters of the same model on exactly the same footing. For the same reason, the mean transit time is not the ratio of simulated stock to simulated input at any date of the simulation, which reflects the imbalance of the soil as well as the model. Second, the calculation collapses a parameter vector of between six and twenty-six numbers into one number with a physical meaning and a unit of years, which allows models that are otherwise very different to be compared on the same axis.

Because a decomposition model responds to climate and to litter chemistry, the mean transit time is not a constant of the model but a property of the model evaluated under a specific condition for these variables. We therefore fix one reference condition and evaluate every model, and every parameter set, under that same condition. The reference climate is the mean climate of the study plots, with a mean annual temperature of 3.4 degrees Celsius, a temperature amplitude of 13.8 degrees and an annual precipitation of 591 millimetres. The reference litter is the mean litter composition of the study plots, normalised to a total of one, which divides the input into 0.766 non-woody litter, 0.219 fine woody litter and 0.014 coarse woody litter, each with its own chemical composition, so that the part of each model describing the effect of the size of woody litter is exercised as it would be in a real simulation. The mean transit times reported here should therefore be read as the mean transit time of each model under such conditions, not as a universal constant of the model, nor as the transit time of the real soil under changing conditions. The comparison between our calibration and the published parameterisation does not depend on the choice of reference, because both sides of the comparison are evaluated at the same reference.

1.11 The error model

Observations are compared with predictions on the logarithmic scale, so that the error is proportional to the size of the stock. The total spread of the error is fixed at 0.800 on the logarithmic scale, about twice the spread of the observed stocks around the national mean of each campaign, 0.39. That spread is the error of an idealised model that reproduces exactly the change of the national mean

from one campaign to the next and knows nothing about individual plots. Such a model would follow the measurements together with their errors, because the measurements are all the information it has. A process-based model adds information that the measurements do not contain, in the litter input, the climate and the representation of decomposition, and where the measurements are in error a process-based model is expected to depart from them. The error term has to leave room for that departure.

Four further reasons place the realistic error above that of the idealised model. The litter input carries errors of its own at each plot: the variation of the litter input between plots, 0.41, is followed only weakly by the observed stocks, and counted as an independent error it raises the total to 0.57. The observed errors have heavier tails than a normal distribution. The observed sample excludes outlying plots. Part of the measurement error, in the stone volume, in the bulk density and in the modelled deep tail of the profile, is shared between plots and does not appear in a spread between plots. A value at the upper end of the plausible range was preferred, because a larger error weakens the pull of the observations relative to the priors, so that results attributed to the observations are stated conservatively.

Observations of soil carbon are not independent of one another. Observations of the same plot share whatever the models get systematically wrong about that plot, observations from the same part of the country share whatever the models get wrong about that part of the country, and observations from the same campaign share whatever the protocol of that campaign did differently. The error of each observation is therefore divided into four parts: a part shared by all observations in the same band of latitude, a part shared by all observations of the same plot, a part shared by all observations of the same campaign, and a part that belongs to the observation alone,

$$\log y_{ij} \;=\; \log f_{ij}(\theta) \;+\; u^R_{r(i)} \;+\; u^P_i \;+\; u^C_j \;+\; e_{ij},$$

where y_{ij} is the stock observed on plot i in campaign j, $f_{ij}(\theta)$ the stock predicted by the model, and u^R, u^P, u^C and e the four parts. Each part is treated as a random error with a mean of zero and a fixed standard deviation, and no part is estimated for any individual plot, band or campaign. Instead, the likelihood is averaged over all the values each shared part could take, weighted by how probable each value is, which for normal errors can be done exactly; the shared parts then disappear from the calculation and leave only a correlation between observations that share a plot, a band or a campaign.

The standard deviations of the shared parts were taken from the observations themselves, after removing the mean of each campaign. The standard deviation of the regional part, 0.117, is the spread between the means of eight bands of latitude, whose boundaries were placed so that each band contains the same number of plots, because the plots are unevenly distributed from south to north and bands of equal width would leave some bands almost empty. The standard deviation of the plot part, 0.396, is measured from how strongly two observations of the same plot vary together. Independent errors add by their variances, that is by the squares of their standard deviations, and not by the standard deviations themselves, so the variances of the four parts add up to the variance of the total, 0.800^2, and the standard deviation of the part that belongs to each observation alone, 0.685, is what remains once the three shared parts are accounted for. Dividing the error in this way does not change how far a single observation may lie from the model, but it changes how much the observations are worth together: observations that share part of their error partly repeat one another, and the 1205 observations carry about as much information on the national mean as 220

independent observations would. Both published calibrations of Yasso assumed the errors of the observations to be independent (Tuomi et al., 2009; Viskari et al., 2022), and the calibration of Yasso20 named the correlation of errors within a time series as an effect still to be included (Viskari et al., 2022). Treating correlated observations as independent leads to parameter uncertainties that are too narrow (Oberpriller et al., 2021), and the shared parts of the error used here are a way of avoiding that.

The standard deviations of the campaign parts cannot be measured from the data, because a shared offset between campaigns and a genuine change of the stock between campaigns look the same in the observations. The standard deviations were therefore prescribed: 0.030 for each of the two later campaigns, which followed similar protocols, and twice that, 0.060, for the first campaign, which differs from the later two in three documented ways. The organic layer of the first campaign excluded the litter and moss layer, which is added here from each plot's measurement of 2006; the fieldwork of the first campaign extended over a decade; and the carbon content of its mineral layers below 5 cm was mostly predicted from the loss of mass on ignition rather than measured directly (Kramarenko, 2012).

1.12 Sampling

Each model was calibrated with five independent runs of 50 000 iterations, of which the first 5 000 were discarded, using the differential evolution Markov chain sampler of ter Braak and Vrugt (2008), in the implementation provided by the `BayesianTools` package (Hartig et al., 2023), version 0.1.9, under R 4.6.1 (R Core Team, 2023). The sampler proposes each new parameter vector from the difference between past states of its chains, occasionally along the line joining the current state to a past one, so that the proposals adapt to the shape of the posterior; this suits posteriors with strongly correlated parameters, such as the one described in this paper. Each run maintains three interacting chains, so that the five runs together provide fifteen chains. Every calibration in this paper, including those of Sections 1.13 and 1.14, uses the same settings.

The information that the observations add to each parameter was measured as the Kullback–Leibler divergence of the marginal posterior from the marginal prior, estimated from kernel densities of the samples. The divergence is zero when the posterior is identical to the prior, that is when the observations say nothing about the parameter, and grows as the observations narrow the distribution or move the distribution away from the prior. The divergence is expressed in nats, the unit that results when information is measured with the natural logarithm, as bits are the unit that results with the logarithm to base two. Two cases give a sense of the scale: a posterior with the same centre as its prior and one quarter of its width corresponds to about 0.9 nats, and a posterior with the same width as its prior but moved by one prior standard deviation corresponds to 0.5 nats. Convergence was assessed with the potential scale reduction factor and with the effective sample size, and the rate of proposals rejected as numerically impossible was monitored throughout. For all eighteen calibrations (six models, each calibrated with the transient initialisation, with a balanced start, and with a balanced start and the climate held), the largest potential scale reduction factor of any parameter was 1.009 and the smallest effective sample size 1716.

1.13 The comparison with a soil started in balance

To test whether the transient initialisation is needed, every model was calibrated a second time with the soil started in balance. The calibration is identical in every respect — data, error model, priors, sampling — except for one factor: the starting state. In 1985 the soil is set to the state at which its gains and losses balance under the litter input of 1985–1989, scaled by σ_{input}, and the reference climate of 1961–1984; σ_{init} is fixed at one, which makes the litter path of Section 1.7 flat, and is removed from the parameters. The prior of σ_{input} is the same as in the transient calibration. The soil then receives the observed litter and climate after 1985, exactly as in the transient calibration.

The balanced start is the steady-state assumption applied at the start of the measurements, and not the procedure of any particular inventory. The test is deliberately the strongest fair version of that assumption: every other parameter is recalibrated, so the models are free to compensate for the missing deficit through the litter input, the transfer fractions and the response to climate. A soil started in balance in 1985 is not expected to stay flat, because the litter input rose by about half until 2006; what distinguishes the two starts is the size of the accumulation they produce, and whether a model started in balance can match the measured rates of accumulation without an inherited deficit.

Four expectations were written down before the calibrations were run: that both starts would reproduce the level of the campaign means; that the balanced start would produce too little accumulation over 1985–2006 and 1985–2024 in most models; that it would compensate by moving along the trade-off between litter input and transit time, with a larger σ_{input}, a shorter transit time or both; and that Yasso07, whose single climate modifier rescales time in all pools at once, would have the most room to compensate. The expectation that would have refuted the need for the transient initialisation was also written down: that the balanced start reproduces the observed rates within their intervals, at no material cost in fit, in most models. [Keep the expectations in the Methods (they make the test falsifiable) and report in the Results which held and which did not (two did not: the balanced start overshoots the stock of the first campaign, and its σ_{input} went down rather than up)? Storyline v4 leaves the choice open.]

The two calibrations are compared on the trajectory of the national mean against the three campaigns, on the rates of change over 1985–2006, 2006–2024 and 1985–2024, on the position of each model along the trade-off between litter input and mean transit time, and on the fit. The log-likelihoods of the two calibrations can be compared directly, because the data, the error model and its spread are the same and the balanced start is a special case of the transient one; the comparison is nevertheless made mainly on the rates, because the information criteria inherit the overconfidence that the error model (Section 1.11) only partly removes.

1.14 The division of the modelled sink

The sink each calibrated model produces over 1985–2024 was divided among three sources: the state inherited from before 1985, the change of the litter input since 1985, and the climate. The division is built in the same way as the comparison of Section 1.13, and on it. One further calibration was made for every model, with the soil started in balance *and* the climate held at its reference, the mean of 1961–1984, in every year after 1985, in the calibration as in the simulation. Each model is thereby calibrated three times on the same measurements, adding one assumption at a time:

1. the transient initialisation, with the observed litter and climate (the calibration used in every

other result);

2. a balanced start, with the observed litter and climate (Section 1.13);
3. a balanced start, with the observed litter and the climate held at the reference.

The three parts of the sink are the differences between the three calibrations, year by year. The inherited state is the sink of the first minus the sink of the second; the climate is the sink of the second minus the sink of the third; and the litter change is the sink of the third, which exists only because the litter departs from its 1985–1989 level, since a soil in balance under an unchanging litter input and climate gains nothing. The three parts add up to the sink of the first calibration by construction.

Because every calibration is fitted to the measurements under its own assumption, each part compares two fitted models that differ by one assumption, and includes the change of the other parameters that the data demand once that assumption is made. The division is therefore a comparison between assumptions, and not the effect of one source on a model held fixed. The climate changes the share of carbon that decomposes each year, so its effect depends on how much carbon is present, and it has a joint effect with the inherited deficit and another with the litter change. In the three parts the joint effects are assigned by the order of the calibrations: the inherited part is measured under the observed climate and contains the joint effect of climate and inheritance, and the climate part is measured on the observed litter and contains the joint effect of climate and litter. [Five terms instead of three: the two joint effects can be reported as terms of their own with two further calibrations (transient start with the climate held; balanced start with the litter held at 1985–1989). Planned with the new input data; rewrite this paragraph then.]

The parts are computed for the plots measured in all three campaigns (310 plots) and for 50 posterior draws of each calibration, and the national values are plain averages over plots and draws, which keep the parts adding up to the sink. Two checks are made for every model: for every draw of the two balanced-start calibrations, the soil with the litter and the climate both held at their references keeps a sink of zero on every plot, and the three parts add up to the sink of the first calibration in every year and draw. The size of the climate part depends on the choice of the reference climate, whereas the ordering of the parts does not; the reference was chosen as the climate the soil had lived in before the observations began, the same that drives the initialisation.

1.15 Simulations beyond the observations

Four analyses extend the calibrated models beyond the observations. They give orders of magnitude for the accumulation that remains and for its sensitivity, and are not forecasts of the national soil carbon balance: they describe the calibrated models on the study plots, per hectare, under stated forcing.

The projection. Each model is run for 60 years after the last observation, with the litter input held at its value in the last observed year, 2024, and with the climate of the last 20 observed years repeated three times, so that the projection contains neither a change of climate nor a change of forest growth. Because the repeated climate alternates between a cooler and a warmer decade, rates of change over the projection are computed over whole 20-year cycles. The projection uses 100 posterior draws of each calibration and the plots measured in all three campaigns, and is made both

for the transient calibration and for the calibration started in balance (Section 1.13), so that the effect of the starting assumption on the projected sink can be read directly.

The distance from balance. The equilibrium stock of each plot is the stock at which gains and losses would balance if the litter input and climate of the projection were held indefinitely, computed with the routine of each model for the state of balance, as for the mean transit time (Section 1.10); the distance from balance is the equilibrium stock divided by the stock of 2024, minus one.

More litter input. The projection is repeated with the litter input increased by 20 per cent from the first projected year onwards. Because the models are linear in their input, the eventual gain of stock under that increase is 20 per cent of the equilibrium stock, and the scenario shows how much of the gain is realised within 60 years. The same linearity gives, with no further simulation, the increase of litter input that would hold a given sink over a given period: for each posterior draw the sink over the period is a straight line in the input multiplier, passing through the projections with the input held and increased by 20 per cent, and the multiplier that gives the target sink is read off that line. The targets are the observed rates of 1985–2024 and 2006–2024, and the periods the first 10 and 20 years of the projection. Because the projection carries no further warming, the multipliers obtained are lower bounds for a warming climate.

Warming. The response to warming is measured by adding a linear rise of temperature to the repeated climate of the projection, from zero in 2025 to its full value in 2084, and comparing the stock in 2084 with that of the projection without the rise. Three scenarios are taken from the projections for Finland of Ruosteenoja and Jylhä (2021), SSP1-2.6, SSP2-4.5 and SSP5-8.5, whose warming of the annual mean temperature is given relative to 1981–2010. Part of that warming has already happened within the climate being repeated: on the study plots the mean of 2005–2024 is 1.0 degree warmer than that of 1985–2010, which stands in for 1981–2010. The full value of the rise is therefore the warming still to come, 1.3, 2.6 and 5.4 degrees for the three scenarios, so that the warming already observed is not counted twice. The litter input is held at its 2024 value under warming, with no response of forest growth, so the rise isolates the response of decomposition and biases every model toward a loss of carbon; the result is a sensitivity, not a projection.

The response to warming is computed for three parameterisations of each version of Yasso: our calibration (40 posterior draws), the calibration with a balanced start (40 posterior draws, Section 1.13), and the published parameterisation. Both arms start from the state built with the run's own pre-run climate (Section 1.7). For Yasso15 and Yasso20 the published arm uses draws from the published posterior distributions of the parameters; for Yasso07, for which no such distribution is available, it uses the published parameter values. The published kinetics are not refitted; only the litter input multiplier of the published arm is set, for each draw, so that the arm reproduces the three campaign means, with σ_{init} held at the median of our calibration. Because the stock is exactly proportional to the litter input multiplier, the adjustment sets the level without changing the kinetics, and the relative response to warming does not depend on it. The balanced-start calibration is run from its own starting state, the steady state under the pre-run climate with $\sigma_{\text{init}} = 1$, and with its own posterior litter input multiplier; comparing its response with that of our calibration shows whether relaxing the steady-state assumption changes the sensitivity of the calibrated models to temperature, and not only their trajectory.

1.16 Predictive evaluation and residual analysis

Predictions were generated for both the calibration plots and the held-out plots by drawing from the posterior and running each draw forward. Model skill is reported as the coefficient of determination and the root mean squared error, separately for the calibration plots and for the held-out plots. Residuals were defined as the logarithm of the observed stock minus the logarithm of the predicted stock, so that a positive residual means that the soil holds more carbon than the model predicts. The residuals of the six models are nearly identical from plot to plot, with a correlation of 0.99 or higher between any two models, and the detailed analyses of the residuals therefore use a single value for each observation, the mean of the residuals of Yasso07 and Yasso15.

Three analyses were made. First, a random forest was fitted to the residuals of each model and evaluated out of bag, with about sixty site variables describing the climate, the texture and chemistry of the soil, the composition of the litter input and the stand in 1985, to measure how much of the residual can be predicted at all and which variables lead. The ranking was checked with a wider set of several hundred variables: the stand records and tree tallies of the four inventories, the chemistry of the soil, topographic wetness, nitrogen deposition, the record of management, and the cover of the understorey vegetation measured in 2023; variables that measure the soil carbon itself were left out. The leading variables of the forest then guided two detailed analyses, made on the mean residual of Yasso07 and Yasso15. Second, the mean residual of each plot was regressed on the basal area of the stand in 1985, fitted as a smooth curve, together with three properties of the site: the fertility of the site, expressed by the forest site type recorded by the inventory; the texture of the mineral soil, recorded in the field in 1985; and the volume share of stones in the soil. Third, the residual of each soil observation was related to the time elapsed since the last regeneration cut before the soil was sampled, a regeneration cut being a felling that removes the stand so that a new one can be established. The year of the cut was taken from the stand records of the inventories of 1985, 1990, 1995 and 2021–2023 and from the forest use declarations filed with the Finnish Forest Centre from 1997 onwards. Where no cut was recorded before the sampling, the time since the stand was established was used instead, taken as the age of the stand in 1985, measured on the tally trees, plus the years elapsed since 1985. Because each plot contributes up to three observations, the comparison between the basal area of 1985 and the time since the cut used a regression with a separate intercept for each plot.

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